Duncan Mak

Software Engineer & Technical Leader
Developer Tools · Programming Languages · Engineering Infrastructure · Developer Productivity

Cambridge, MA 02139
duncanmak@gmail.com

Employment

Oct 2024 ~ May 2026

Microsoft (contract)

Senior Software Engineer · Azure Studio (Design)

Product launch recovery: Led a two-engineer team to recover a delayed Next.js documentation application after the original team departed. Diagnosed architectural and performance problems and brought the project to launch readiness in seven weeks.

180× faster page loads: Eliminated repeated Figma API requests by introducing SQLite-backed caching for JSON and SVG assets, then applied Brotli compression to reduce page load times from approximately three minutes to under one second.

Deployment architecture: Owned Azure deployment for the team's new development effort. Designed a workflow using reusable ARM/Bicep templates and Azure Pipelines to provision independent, database-backed Next.js applications, enabling multiple customers to run isolated installations with separate infrastructure and billing.

Security compliance: Served as the team's security compliance DRI, coordinating with product management and security stakeholders to manage requirements, remediation work, and compliance deadlines.

AI-assisted UI generation: Co-developed a Hackweek 2025 prototype using Model Context Protocol (MCP) to expose design-system documentation to LLMs, enabling natural-language generation of JavaScript interfaces built from the team's component library.

Jul 2023 ~ Jul 2024

Elemental Cognition Inc.

Senior Software Engineer working on the Cogent AI Platform

AI platform productization: Worked in a three-engineer team to bring the Cogent AI research platform to market as a developer product.

Web IDE: Helped redesign and rebuild the Cogent IDE using React, TypeScript, and Tailwind CSS, delivering a polished developer experience in three months.

Compiler and SDK: Extended the Cogent compiler and middleware in Scala and developed its TypeScript SDK, including CommonJS/ESM packaging and API support.

Jul 2020 ~ Feb 2023

Microsoft Inc.

Senior Software Engineer for docs.microsoft.com and learn.microsoft.com

Developer tooling modernization: Led an initiative to adopt a buildless JavaScript development environment using Deno, Lit, and Web Components, reducing the developer feedback loop from over five minutes to seconds.

Engineering leadership: Led a four-engineer squad to improve test coverage, deployment reliability, and internal security compliance for the C#-based Video on Demand service. Maintained the backend and fixed frontend issues using TypeScript and Web Components.

Frontend simplification: Refactored internal administration tools in TypeScript, reducing the codebase to one-fifth of its original size.

Product delivery: Shipped the first version of Microsoft Learn's Organizational Reporting feature within three months of joining the organization.

Jul 2019 ~ Jul 2020

Senior Software Engineer in VS Engineering

Modernize VS Build: Revived an MSBuild modernization initiative stalled for over a year, integrating MSBuild 16 and enabling the XAML Designer team to ship cross-target projects in Visual Studio.

Mac Notarization: Coordinated a cross-company effort to support Notarization on macOS Catalina. Worked in conjunction with Edge, ESRP, Java, .NET, Office and VSCode to ensure Microsoft can ship Mac products for the new macOS release. Visual Studio for Mac shipped on Catalina with day-one support when Notarization began to be enforced.

Jul 2016 ~ Jul 2019

Senior Software Engineering Manager

Engineering leadership: Managed and grew a seven-person engineering infrastructure team responsible for build and release systems supporting the Mono runtime and Visual Studio for Mac.

Collaboration across divisions: Represented DevDiv and collaborated with Office Mac engineering to prepare for migrating Mac build labs from on-premise to hosted inside data centers.

Monitor VSMac build times: Built tooling to monitor the CI performance of Visual Studio for Mac. Using data generated by the tool, identified trends and specific commits that led to increased build times. This work was presented to senior leadership.

Developer workflow tooling: Designed and implemented a ChatOps system using Azure Functions to work with GitHub and Slack APIs using F#, improved the code for the Update server for Visual Studio for Mac in C# and built many tools to create a friction-free engineering system for the engineers working on the Xamarin Platform.

Nov 2011 ~ Jul 2016

Xamarin Inc. (acquired by Microsoft)

Engineering Lead for Release Engineering

First Release Engineer: One of the earliest members to join Xamarin. The first full-time employee responsible for release engineering of the Xamarin Platform.

Build lab: Grew the on-premise Mac build lab from 5 machines to nearly 100 machines, serving a fully remote, global engineering team that grew from 10 members to over 100.

Hiring: Hired and grew the release engineering team from 1 to 4 full time members.

Faster builds on multiple architectures: Using data from our CI server, analyzed build logs and improved the build times for Mono by reducing

unnecessary targets in the Mono makefiles.

Dashboards for optimized infrastructure: Built a dashboard to visualize CI usage and turn off unnecessary jobs, a dashboard to track and display the status of Mac minis in the growing lab. Chose to use Typescript and React, which were both new technologies at the time.

May 2011 ~ Oct 2011

MyEnergy Inc. (acquired by Nest Labs)

Software Engineer

Wrote tools to extract textual information from PDFs using JRuby and Scala 2.8.

Sept 2008 ~ Apr 2011

Center for Brain Science, Harvard University

Software Engineer for the Connectome project

As the only software engineer in the lab, I created and maintained the software stack used by Harvard researchers. I learned to assess, design, and implement software solutions that meet the ever-shifting needs of cutting-edge science research.

Skills

Programming Languages: TypeScript, JavaScript, C#, Scala, F#, Clojure, Scheme, Java, Ruby, Smalltalk

Developer Tools & Language Systems: Compilers, IDEs, SDK development, build systems, developer productivity tooling

Web & Application Development: React, Next.js, Lit, Web Components, HTML/CSS, MCP integration

Cloud & Infrastructure: Azure, ARM/Bicep, Azure Pipelines, CI/CD, infrastructure as code, SQLite

Spoken Languages: English, Cantonese, Mandarin, French, Spanish, Portuguese, Japanese

Education

Jan 2009

Northeastern University Boston, MA
B.A. in Computer Science

Jan 2007

Northeastern University Boston, MA
B.A. in Linguistics